

RECONSTRUCTING MODIS LST PRODUCTS OVER TIBETAN PLATEAU BASED ON RANDOM FOREST

Yuan Cheng¹, Yuxia Li¹, Huanping Wu², Fan Li¹, Yuzhen Li,³ Lei He⁴

1. School of Automation Engineering, University of Electronic Science and Technology of China, Chengdu 611731, China. Email: shyabb@163.com
2. National Climate Center, China Meteorological Administration, Beijing, 100081, China.
3. ChengDu Software Industry Development Center, Chengdu 610041, China.
4. School of Software Engineering, Chengdu University of Information Technology, Chengdu 610025, China.

Abstract

Land Surface Temperature (LST) is an indicator of the thermal condition at the ground-atmosphere interface. The MODIS LST products from Terra and Aqua satellites provide spatially continuous monitoring ground temperature. But it still has limitations at local scale due to atmospheric disturbances and the spatiotemporal heterogeneity of land surface. In this study, a novel algorithm based on Random Forest (RF) is proposed to reconstruct MODIS LST. The RF was trained using MOD11A1, MOD09A1 and MOD15A2 from Terra MODIS and digital elevation data from ASTER DEM as inputs. LST observations from The Tibetan Plateau Soil Moisture and Temperature Monitoring Network (TP-SMTMN) are used as target data. Experimental results indicate the algorithm can improve the estimation MODIS LST products from the aspects of accuracy and data availability.

Index Terms—Land Surface Temperature, MODIS, the Tibetan Plateau, Random Forest

1. INTRODUCTION

Land Surface Temperature (LST) is one of the key variables in the physics of land surface processes. Surface meteorological stations can precisely observe the surface temperature, but it is difficult to obtain spatially continuous and reliable LST data via ground surveys. Remote sensing provides effective means for regional-scale monitoring of LST. MODIS provides LST products such as MOD11L2, MOD11A1, MOD11A2 with accuracy better than 1 K in the range from 263 to 322 K [1]. However, MODIS LST products also have its disadvantages. First, MODIS sensors can only cross four times a day for one area, and it is impossible to obtain observations for the whole day or a specified time because of the fixed transit time. Second, the accuracy of MODIS LST products in different regions also varies due to

the heterogeneity of climate and terrain. Third, MODIS is often disturbed by atmospheric condition such as cloud and water vapor, observations may be the cloud top temperature rather than land surface temperature. For the above reasons, more and more researches have been devoted to the reconstruction of MODIS surface temperature products in recent years. Huang et al [2] used ensemble Kalman filter and Common Land Model version 3.0 (CoLM) to reconstruct MODIS LST products, and the estimation of soil temperature profile has been improved about 1 K. Neteler et al [3] present a self-contained LST reconstruction algorithm based on the use of several histograms and temperature gradient-based outlier filters, reconstruct missing MODIS data using volume spline interpolation.

In this study, we developed the Random Forest (RF) algorithm to reconstruct the MODIS LST products. The RF is trained with near-infrared band reflectivity and Normalized difference vegetation index (NDVI) retrieved from MOD09A1, daytime and nighttime LST from MOD11A1, leaf area index (LAI) from MOD15A2 and digital elevation data from Aster-GDEM as inputs, and with expected outputs from The Tibetan Plateau Soil Moisture and Temperature Monitoring Network (TP-SMTMN). Compared with the MODIS products, the reconstructed LST has the following advantages:

- (1) The accuracy has been improved by reducing the errors caused by the terrain and atmospheric disturbances.
- (2) The LST can be mapped continuously in time by using target data with different acquisition time.
- (3) Reliable estimates will be generated even when some data from MODIS LST products is invalid.

2. DATASETS

2.1 In situ measurements

The Tibetan Plateau Soil Moisture and Temperature Monitoring Network (TP-SMTMN) consists of three

networks with different climatic conditions, namely Maqu, Naqu and Ngari [4]. The Maqu network has been set up on the north-eastern fringe of the Tibetan Plateau (33.5° - 34.25°N, 101.63°-102.75°E), was selected as the study area in this paper. It consists of 20 meteorological stations and

covers an area of approximately $40 \times 80 \text{ km}^2$ which is shown in Fig.1 [5]. Each station in the network collects soil moisture and temperature data every 15 minute. Measurements of land surface temperature at 10:00 AM in Maqu during the period of May 2008 to Oct 2015 are pooled as the target data.

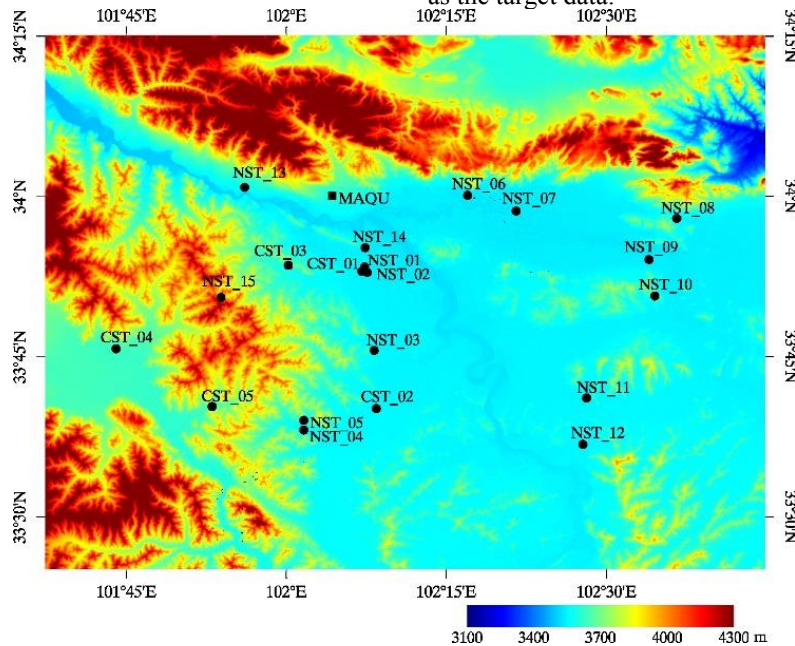


Fig.1 The locations of the 20 monitoring stations of Maqu network [5]

2.2 MODIS products

MODIS is a moderate-resolution imaging spectrometer carried on Terra satellites and Aqua satellites. It has 36 spectral channels and is distributed in the electromagnetic spectrum range of 0.4 to 0.14 μm . The MODIS products used in this study are MOD11A1 (1km surface temperature & emissivity daily product), MOD09A1 (500m surface reflectance 8-day product) and MOD15A2 (500m Leaf Area Index & FPAR 8-day product). Each product uses a sinusoidal projection with a coverage range of about 1100KM*1100KM. Only images collected during the period from 2008 to 2015 were used. The MODIS products were downloaded directly from the Land Processes Distributed Active Archive Center (LPDAAC).

2.3 Auxiliary datasets

ASTER DEM acquired by the ASTER sensor onboard the Terra satellite was collected to present the terrain of the study area. It covers all land areas between 83° north latitude and 83° south latitude, reaching 99% of the earth's land surface. The spatial resolution of ASTER DEM is 30m and can be freely downloaded from the land processes distributed active archive center (<https://lpdaac.usgs.gov/>).

2.4 Data preprocessing

The MODIS data used in this study have been corrected for atmospheric and aerosol effects. All satellite data should be projected to the WGS84 coordinate system and resampled to 1000m spatial resolution. The preprocessing can be completed in batches using the MRT (MODIS Reprojection Tool) tool and ENVI5.3. After removing invalid data, it is necessary to synchronize with the remote sensing data according to the latitude, longitude and acquired time. Considering the disturbances of snow and frozen soil, data only from May to October each year has been used in this paper.

3. MODIS LST RECONSTRUCTION SCHEME

3.1 Influencing factors of LST

Numerous parameters have been successfully used in LST estimation or reconstruction. In this study, 5 LST related indices, NDVI, LAI, near-infrared band reflectivity and elevation, were used to reconstruct MODIS LST products. Among them, NDVI and LAI are used here to parameterize the vegetation coverage [6]. Near-infrared band reflectivity is used to parameterize the influence of solar radiation. And the elevation is used to reduce the effects of surface characteristics and the location [7]. Land surface temperature is also influenced by soil moisture and precipitation [8]. Considering the coarse spatial resolutions and acquisition

difficulty of these two factors, they are not considered in this study.

3.2 Random Forest method

Ensemble learning has been widely used in classification and regression problems. The Random Forest (RF) is one of the ensemble methods based on decision trees and can reduce the risk of overfitting by averaging these trees [9]. The schematic diagram of the RF algorithm is shown in Fig.2. First, it performed sampling with bootstrap in the original training data set D . After m times of sampling, a new training set D_1 with m samples can be obtained. T-sampling is performed in the same way, and n input features $\{X_1, X_2 \dots X_n\}$ are randomly selected in each round of sampling, that is, samples (rows) and input features (columns) are randomly selected from the original training set D , and finally T new datasets $\{D_1, D_2 \dots D_T\}$ can be obtained. Based on each new data set, T "weak learners" $\{h_1, h_2 \dots h_T\}$ can be trained, and finally the results of each "weak learner" are combined by voting or averaging. In the random forest model, the existence of sample perturbations (random row sampling) and attribute perturbations (random column sampling) makes each decision tree trained different, ensuring the diversity of the model. If the decision trees are combined according to the average method, the final output result is shown in Eq. (1).

$$H(x) = \frac{1}{T} \sum_{i=1}^T h_i(x) \quad (1)$$

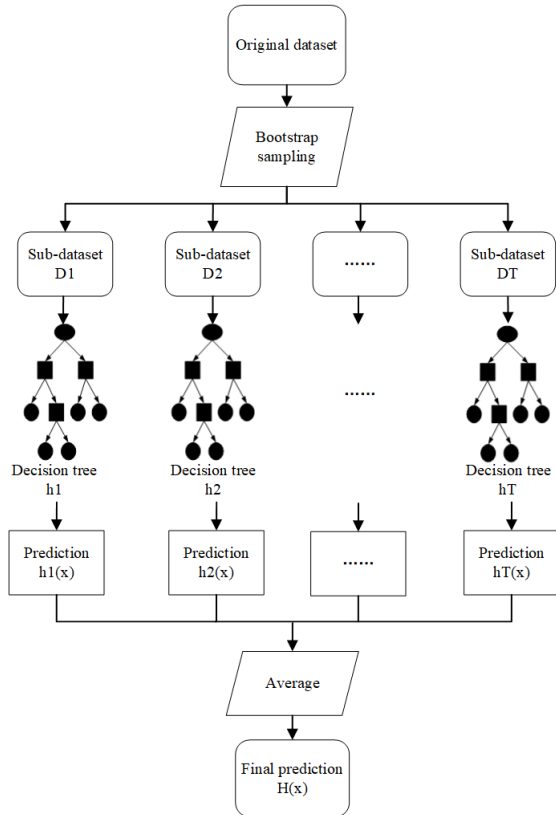


Fig.2 Schematic diagram of the random forest algorithm.

In this paper, MOD09A1, MOD11A1, MOD15A2 and Aster-GDEM are used as source data. The random forest is trained with near-infrared band reflectivity, Normalized difference vegetation index (NDVI), leaf area index (LAI), daytime and nighttime LST and digital elevation as inputs, and the observed surface temperature data at 10:00 am in the TP-SMTMN network is used as the desired output. All the MODIS products used in the current study are quality controlled, only cloud-free MOD09A1 images were acquired and analyzed. After quality control, a total of 16615 data points during the years 2008 to 2014 are obtained, 80% of the data is used for training, 20 % of the data is used to validate the accuracy of the model. In order to verify the time migration of the RF-based reconstructed algorithm, we used a total of 1796 data points of years 2015 to evaluate the accuracy of the algorithm.

4. RESULTS AND ANALYSIS

The root-mean-square error (RMSE), the mean absolute error (MAE) and the coefficient of determination (r) are used for results evaluation in this paper. These parameters were calculated as follows:

$$RMSE = \sqrt{\frac{1}{N} \sum_{i=1}^N (f_i - y_i)^2} \quad (2)$$

$$MAE = \frac{1}{N} \sum_{i=1}^N |f_i - y_i| \quad (3)$$

$$R = \frac{Cov(F,Y)}{\sigma_F \sigma_Y} \quad (4)$$

Where N is the number of samples, F is the sequence of predicted values, Y is the sequence of true values.

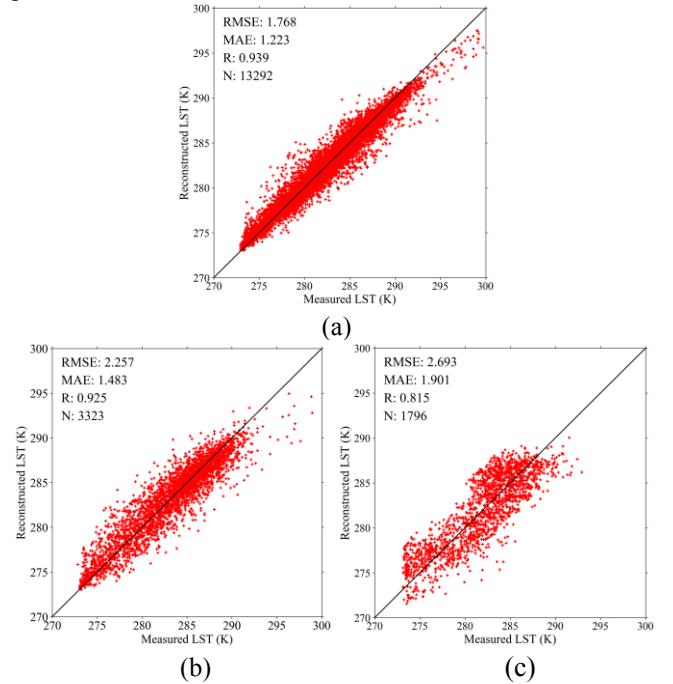


Fig.3 Relationships between the reconstructed LST and measured LST on (a) training dataset; (b) validating dataset; (c) testing dataset.

Fig.3 shows the relationship between the reconstructed LST using RF algorithm and measured LST. For the training dataset, the RMSE, MAE and R of the RF estimations were 1.768 K, 1.223 K and 0.939, respectively. For the validating dataset, the RMSE, MAE and r were 2.257 K, 1.483 K and 0.925, respectively. For the testing dataset, the RMSE, MAE and r were 2.693 K, 1.901 K and 0.815, respectively. The average land surface temperature value of Maqu network can be obtained through the meteorological stations. Fig.4 shows temporal evolution of the in-situ and reconstructed LST in Maqu network from April to October 2015. Areal-averaged reconstructed LST agrees well with the measured LST. There is a high correlation ($R = 0.913$) between the reconstructed LST and the measured LST. The accuracy at the network scale (RMSE = 1.618 K, MAE = 1.336 K) is even better than it is at the point scale. This is because some of the errors can be eliminated after averaging the measured LST values of all meteorological stations. In general, the RF-based method is practical and effective to improve the estimation of land surface temperature.

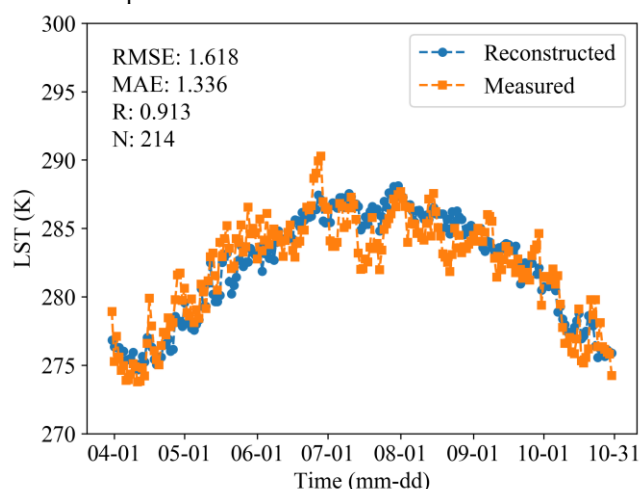


Fig.4 Time series comparison of reconstructed LST against measured LST in Maqu network from April to October 2015

5. CONCLUSIONS

This paper proposed an empirical algorithm based on Random Forest (RF) to reconstruct MODIS land surface temperature products. The algorithm performed well both on validation datasets and test datasets. We have improved the precision of LST measurements and derived LST estimates between MODIS measurement intervals. However, considering the differences in climatic conditions at the three networks of TP-SMTMN, data from the Naqu and Ngari should also be introduced to further improve the accuracy of the algorithm.

6. ACKNOWLEDGEMENTS

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